Chapter 2: Axioms of Probability

2.1 Introduction

This chapter introduces the concept of the probability of an event, starting with the definitions of sample space and events, and then establishing the axioms that govern probability theory.

2.2 Sample Space and Events

Definitions

- **Sample Space (S):** The set of all possible outcomes of an experiment.
  - *Example 1 (Child's Sex):* $S = \{g, b\}$ where g is girl, b is boy.
  - *Example 2 (Race):* For 7 horses, $S = \{\text{all } 7! \text{ permutations of } (1, ..., 7)\}$.
  - *Example 3 (Two Coins):* $S = \{(H, H), (H, T), (T, H), (T, T)\}$.
  - *Example 4 (Two Dice):* $S = \{(i, j) : i, j = 1, ..., 6\}$ (36 outcomes).
  - *Example 5 (Transistor Lifetime):* $S = \{x : 0 \leq x < \infty\}$ (Continuous).
- **Event (E):** Any subset of the sample space S. An event is said to have **occurred** if the outcome of the experiment is contained in E.
  - *Example:* In the dice experiment, if $E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, then E is the event that the sum of the dice is 7.

Set Operations on Events

- **Union ($E \cup F$):** Consists of outcomes that are in E **or** in F **or** in both. Occurs if *either* E or F occurs.
- **Intersection (EF or $E \cap F$):** Consists of outcomes that are in **both** E and F.
- **Mutually Exclusive:** Two events E and F are mutually exclusive if $EF = \emptyset$ (the null event, containing no outcomes). They cannot both occur.
- **Complement (E^c):** Consists of all outcomes in S that are **not** in E. E^c occurs if and only if E does not occur. Note: $S^c = \emptyset$.
- **Subset ($E \subset F$):** If all outcomes in E are also in F. The occurrence of E implies the occurrence of F.

Algebraic Rules for Events

- **Commutative Laws:** $E \cup F = F \cup E$; $EF = FE$
- **Associative Laws:** $(E \cup F) \cup G = E \cup (F \cup G)$; $(EF)G = E(FG)$
- **Distributive Laws:** $(E \cup F)G = EG \cup FG$; $EF \cup G = (E \cup G)(F \cup G)$

De Morgan’s Laws

Useful for relating unions and intersections of complements:

1. $(\bigcup_{i=1}^{n} E_i)^c = \bigcap_{i=1}^{n} E_i^c$
2. $(\bigcap_{i=1}^{n} E_i)^c = \bigcup_{i=1}^{n} E_i^c$

Proof Logic: An outcome is in the complement of the union iff it is not in *any* of the events (intersection of complements). Conversely, an outcome is in the complement of the intersection iff it is not in *all* events (union of complements).

2.3 Axioms of Probability

Frequency Definition vs. Axiomatic Approach

- **Relative Frequency:** $P(E) = \lim_{n \to \infty} \frac{n(E)}{n}$, where $n(E)$ is the number of occurrences of E in n repetitions.
  - *Drawback:* We cannot prove this limit exists or is constant.
- **Axiomatic Approach:** We assume probabilities exist and satisfy specific axioms consistent with intuition.

The Three Axioms

Consider an experiment with sample space S. For each event E, there exists a number $P(E)$ such that:

- **Axiom 1:** $0 \leq P(E) \leq 1$
- **Axiom 2:** $P(S) = 1$
- **Axiom 3:** For any sequence of mutually exclusive events $E_1, E_2, \ldots$ (where $E_i E_j = \emptyset$ for $i \neq j$):

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

Implications

- $P(\emptyset) = 0$.
- For a finite sequence of mutually exclusive events: $P(\cup_{i=1}^{n} E_i) = \sum_{i=1}^{n} P(E_i)$.

2.4 Some Simple Propositions

- **Proposition 4.1:** $P(E^c) = 1 - P(E)$
  - *Proof:* Since $S = E \cup E^c$ and they are disjoint, $1 = P(S) = P(E) + P(E^c)$.
- **Proposition 4.2:** If $E \subset F$, then $P(E) \leq P(F)$.
  - *Proof:* $F = E \cup E^c F$. Thus $P(F) = P(E) + P(E^c F)$. Since $P(E^c F) \geq 0$, the result follows.
- **Proposition 4.3:** $P(E \cup F) = P(E) + P(F) - P(EF)$
  - *Reasoning:* Adding $P(E)$ and $P(F)$ counts the intersection $P(EF)$ twice, so we subtract it once.
- **Proposition 4.4 (Inclusion-Exclusion Identity):**

$$P(E_1 \cup E_2 \cup \cdots \cup E_n) = \sum P(E_i) - \sum\sum P(E_i E_j) + \sum\sum\sum P(E_i E_j E_k) - \cdots + (-1)^{n+1} P(E_1 \cdots E_n)$$

  - *Summary:* Sum of singles - Sum of pairs + Sum of triples - ...

Bonferroni Inequalities

- $P(\cup E_i) \leq \sum P(E_i)$
- $P(\cup E_i) \geq \sum P(E_i) - \sum_{j<i} P(E_i E_j)$
- Truncating the inclusion-exclusion series at any term gives an upper bound (if stopping after a positive term) or a lower bound (if stopping after a negative term).

2.5 Sample Spaces Having Equally Likely Outcomes

If $S = \{1, 2, \ldots, N\}$ and all outcomes are equally likely, then $P(\{i\}) = \frac{1}{N}$.
For any event E:

$$P(E) = \frac{\text{Number of outcomes in } E}{\text{Number of outcomes in } S}$$

Key Examples

1. **Dice Sum:** Probability sum of two dice is 7.
   - Outcomes in E: $(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$ (6 outcomes).
   - Total outcomes: 36.
   - $P(E) = 6/36 = 1/6$.
2. **Committee Selection:** Choosing 5 people from 6 men and 9 women. Probability of 3 men and 2 women.
   - Total committees: $\binom{15}{5}$.
   - Committees with 3m, 2w: $\binom{6}{3}\binom{9}{2}$.
   - $P = \frac{\binom{6}{3}\binom{9}{2}}{\binom{15}{5}} = \frac{240}{1001}$.
3. **Poker Hands:**
   - **Straight:** 5 cards, consecutive values, not all same suit.
     - Total hands: $\binom{52}{5}$.
     - Outcomes: 10×4^5 (possible sequences $\times$ suit choices) minus 40 (straight flushes).
     - $P \approx 0.0039$.
   - **Full House:** 3 of one rank, 2 of another.

- Outcomes: $13 \times \binom{4}{3} \times 12 \times \binom{4}{2}$.
- $P \approx 0.0014$.

4. **Birthday Problem:** Probability that in a room of n people, no two share a birthday.
   - $P(\text{no match}) = \frac{365 \times 364 \times \cdots \times (365-n+1)}{(365)^n}$.
   - Surprising result: For $n \geq 23$, probability of a match > 0.5.
5. **The Matching Problem:** N men throw hats in center and pick randomly. Probability *no one* picks their own hat.
   - Uses Inclusion-Exclusion Principle.
   - $P(\text{no match}) = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^N \frac{1}{N!}$.
   - For large N, this approaches $e^{-1} \approx 0.368$.
6. **Runs:** Independent trials of wins/losses. A "run" is a consecutive sequence of identical outcomes.
   - If n wins and m losses are arranged randomly, probability of r runs of wins:
   - $P(r \text{ runs of wins}) = \frac{\binom{m+1}{r}\binom{n-1}{r-1}}{\binom{n+m}{n}}$.

2.6 Probability as a Continuous Set Function

Sequences of Events

- **Increasing Sequence:** $E_1 \subset E_2 \subset \cdots \subset E_n \subset \cdots$.
  - $\lim_{n\to\infty} E_n = \bigcup_{i=1}^{\infty} E_i$.
- **Decreasing Sequence:** $E_1 \supset E_2 \supset \cdots \supset E_n \supset \cdots$.
  - $\lim_{n\to\infty} E_n = \bigcap_{i=1}^{\infty} E_i$.

Proposition 6.1

If $\{E_n, n \geq 1\}$ is either an increasing or decreasing sequence:

$$\lim_{n\to\infty} P(E_n) = P\left(\lim_{n\to\infty} E_n\right)$$

Example: The Infinite Urn Paradox

- **Experiment 1:** At $1/2^n$ mins to 12PM, add balls $10n-9$ to $10n$, remove ball $10n$.
  - Balls $1-9$, $11-19$, etc., never removed. Infinite balls remain at 12PM.
- **Experiment 2:** Add 10 balls, remove ball n at step n.
  - Ball n is removed at step n. Urn is **empty** at 12PM.
- **Experiment 3:** Add 10 balls, remove *random* ball.
  - With probability 1, the urn is empty at 12PM. (Proof uses decreasing sequence of events E_n = ball 1 is still in urn after n steps).

2.7 Probability as a Measure of Belief

- **Subjective Probability:** Interpreting probability as a measure of an individual's degree of belief in a statement (e.g., "90% chance Shakespeare wrote Hamlet").
- **Consistency:** Subjective probabilities should still satisfy the three axioms of probability to be logically consistent. If they do not (e.g., $P(\text{rain}) + P(\text{no rain}) \neq 1$), they are invalid.

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